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PRESS DIE ASSEMBLY

Abstract

A press die assembly includes a fixed die and a movable die. The fixed die includes a fixed-side forming section that includes a first forming surface, and multiple metal fixed-side support blocks arranged in a planar direction. The fixed-side support blocks support the fixed-side forming section from a side opposite to the first forming surface. The movable die includes a movable-side forming section that includes a second forming surface that faces the first forming surface, and multiple metal movable-side support blocks arranged in the planar direction. The movable-side support blocks support the movable-side forming section from a side opposite to the second forming surface. The Young's moduli of the fixed-side support blocks increase with increasing proximity to a central portion of the first forming surface. The Young's moduli of the movable-side support blocks increase with increasing proximity to a central portion of the second forming surface.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-033751, filed on Mar. 6, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] The present disclosure relates to a press die assembly.

2. Description of Related Art

[0003] Japanese Laid-Open Patent Publication No. 2015-131344 discloses a press die assembly that includes a lower die and an upper die, which are configured to press a sheet material. The forming surfaces of the lower die and the upper die are formed by segments that are separated from each other. The segments of the lower die and the upper die are supported on a base using shims.

[0004] In the press die assembly disclosed in the above-described publication, the central portion of each forming surface is more likely to be deformed by the pressure during press forming than the peripheral portion. In such cases, insufficient pressing pressure is applied to the central portion of the workpiece, resulting in an uneven distribution of pressing pressure on the workpiece. Consequently, the forming accuracy of the press die assembly may be adversely affected.

SUMMARY

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0006] In one general aspect, a press die assembly includes a fixed die and a movable die configured to move toward and away from the fixed die. The press die assembly is configured to press a metal sheet. The fixed die includes a fixed-side forming section that includes a first forming surface, and multiple metal fixed-side support blocks arranged in a planar direction that is orthogonal to a movement direction of the movable die. The fixed-side support blocks support the fixed-side forming section from a side opposite to the first forming surface in the movement direction. The movable die includes a movable-side forming section that includes a second forming surface that faces the first forming surface, and multiple metal movable-side support blocks arranged in the planar direction. The movable-side support blocks support the movable-side forming section from a side opposite to the second forming surface in the movement direction. Young's moduli of the fixed-side support blocks increase with increasing proximity to a central portion of the first forming surface in the planar direction. Young's moduli of the movable-side support blocks increase with increasing proximity to a central portion of the second forming surface in the planar direction.

[0007] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a plan view of a separator for a fuel cell that is manufactured using a press die

assembly according to an embodiment.
[0009] FIG. **2** is a cross-sectional view taken along line **2-2** in FIG. **1**.
[0010] FIG. **3** is a cross-sectional view of a press die assembly according to the embodiment.
[0011] FIG. **4** is a plan view of the fixed-side support block shown in FIG. **3**.
[0012] FIG. **5** is a plan view showing a positional relationship between the fixed-side support block shown in FIG. **3** and a first forming surface.
[0013] FIG. **6** is a plan view showing a positional relationship between the movable-side support block shown in FIG. **3** and a second forming surface.
[0014] FIG. **7** is a plan view showing a positional relationship between a fixed-side support block and a first forming surface according to a first modification.
[0015] FIG. **8** is a plan view showing a positional relationship between a fixed-side support block and a first forming surface according to a second modification.
[0016] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0017] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.
[0018] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.
[0019] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”
[0020] A press die assembly **10** according to an embodiment will now be described with reference to FIGS. **1** to **6**.
[0021] The press die assembly **10** produces a separator **100** for a fuel cell by pressing a metal sheet M.
[0022] First, the structure of the separator **100** will be described.

Separator **100**

[0023] As shown in FIG. **1**, the separator **100** has the shape of a rectangular sheet with long sides and short sides.
[0024] In the following description, the long-side direction of the separator **100** will be referred to as a first direction X, and a direction orthogonal to both the first direction X and the thickness direction of the separator **100** will be referred to as a second direction Y. The second direction Y agrees with the short-side direction of the separator **100**.
[0025] The material of the separator **100** is, for example, a metal material such as stainless steel or titanium.
[0026] The separator **100** includes three manifolds **101** at each of the opposite ends in the first direction X. Fluid is suppled or discharged through the manifolds **101**. The manifolds **101** extend through the separator **100** in the thickness direction. The fluid is, for example, any one of fuel gas such as hydrogen, oxidant gas such as air, and coolant.
[0027] As shown in FIG. **2**, the separator **100** includes groove-shaped flow passages **102** on the opposite surfaces in the thickness direction. Fluid flows through the flow passages **102**. The flow passages **102** are provided in parallel in the second direction Y. The flow passages **102** allow fluid to flow from a manifold **101** at one end in the first direction X of the separator **100** to a manifold **101** at the other end of the separator **100**.

[0028] A rib **103** is provided between any two adjacent flow passages **102**. The ribs **103** extend along the flow passages **102** and define the flow passages **102**. The flow passages **102** that are provided on one side of the separator **100** act as ribs **103** on the other side of the separator **100**. The ribs **103** that define the flow passages **102** on one side of the separator **100** act as flow passages **102** on the other side of the separator **100**.

Press Die Assembly **10**

[0029] As shown in FIG. **3**, the press die assembly **10** includes a fixed die **20** and a movable die **30** configured to move toward and away from the fixed die **20**. The press die assembly **10** uses the fixed die **20** and the movable die **30** to press a metal sheet M, which is a base material of the separator **100**, so as to form multiple flow passages **102** on the surfaces of the metal sheet M.

[0030] Hereinafter, the direction in which the movable die **30** moves toward and away from the fixed die **20** will simply be referred to as a movement direction, and a planar direction orthogonal to the movement direction will simply be referred to as a planar direction.

Fixed Die **20**

[0031] The fixed die **20** includes a press bed **21**, a fixed-side shoe **22**, and a fixed-side backing plate **23**. The fixed-side shoe **22** is fixed to the upper surface of the press bed **21**. The fixed backing plate **23** is fixed to the upper surface of the fixed-side shoe **22**. The press bed **21**, the fixed-side shoe **22**, and the fixed-side backing plate **23** are made of metal.

[0032] The fixed die **20** includes a fixed-side holding plate **24**, fixed-side support blocks **25**, shims **26**, and fixed-side forming section **27**. The fixed-side holding plate **24** is fixed to the upper surface of the fixed-side backing plate **23**. The fixed-side holding plate **24** has a holding hole **24***a*, which extends vertically through the fixed-side holding plate **24**. The fixed-side support blocks **25**, the shims **26**, and the fixed-side forming section **27** are held in the holding hole **24***a*. The fixed-side holding plate **24**, the fixed-side support blocks **25**, the shims **26**, and the fixed-side forming section **27** are made of metal.

[0033] The fixed-side forming section **27** includes a first forming surface **27***a* that forms the metal sheet M. The first forming surface **27***a* has protrusions and recesses for forming the flow passages **102** on the metal sheet M. The fixed-side forming section **27** is formed by fixed-side forming blocks **28**, which are arranged side by side while being adjacent to each other in the planar direction. The first forming surface **27***a* is formed by continuous upper surfaces of the fixed-side forming blocks **28**.

[0034] The fixed-side support blocks **25** are arranged side by side on the fixed-side backing plate **23** while being adjacent to each other in the planar direction. Each fixed-side support block **25** supports one of the fixed-side forming blocks **28** from a side opposite to the first forming surface **27***a* with the corresponding shim **26** in between. Each fixed-side support block **25** has the same shape as the fixed-side forming block **28** supported by the fixed-side support block **25** in plan view, and has a size smaller than or equal to that of the fixed-side forming block **28**. Therefore, the arrangement of the fixed-side support blocks **25** in plan view is the same as the arrangement of the fixed-side forming blocks **28** in plan view.

[0035] As shown in FIGS. **4** and **5**, the fixed-side support blocks **25** are arranged in the long-side direction of the metal sheet M, that is, in the first direction X during press forming. The fixed-side support blocks **25** include one fixed-side support block **25**A, two fixed-side support blocks **25**B, two fixed-side support blocks **25**C, and two fixed-side support blocks **25**D.

[0036] As shown in FIG. **3**, the fixed-side support block **25**A supports the fixed-side forming block **28** that forms a central portion of the first forming surface **27***a*. The fixed-side support blocks **25**B are adjacent to the opposite sides of the fixed-side support block **25**A in the first direction X. Each fixed-side support block **25**C is adjacent to a side of the corresponding fixed-side support block **25**B that is opposite to the fixed-side support block **25**A.

[0037] As shown in FIG. **4**, the fixed-side support blocks **25**D are adjacent to the opposite sides in the second direction Y of the fixed-side support block **25**A and the fixed-side support blocks **25**B.

The opposite ends of each fixed-side support block **25**D in the first direction X are adjacent to the fixed-side support blocks **25**C in the first direction X.

[0038] As shown in FIGS. **3** and **5**, a portion of the fixed-side forming section **27** that forms the flow passages **102** includes the central portion of the first forming surface **27***a*, and is formed by the fixed-side forming blocks **28**. The portion of the fixed-side forming section **27** that forms the flow passages **102** is supported by the fixed-side support blocks **25**A, **25**B, and **25**C.

[0039] The Young's moduli of the fixed-side support blocks **25** differ from each other. Specifically, the Young's moduli of the fixed-side support blocks **25** increase with increasing proximity to the central portion of the first forming surface **27***a* in the planar direction. In the present embodiment, among the fixed-side support blocks **25**, the fixed-side support block **25**A, which is closest to the central portion of the first forming surface **27***a*, has the highest Young's modulus. The Young's modulus of the fixed-side support blocks **25**B is lower than the Young's modulus of the fixed-side support block **25**A and higher than the Young's modulus of the fixed-side support blocks **25**C. The Young's modulus of the fixed-side support blocks **25**C is the same as the Young's modulus of the fixed-side support blocks **25**D.

[0040] The difference in Young's modulus between the fixed-side support blocks **25** is achieved by differences in the materials. The fixed-side support block **25**A is formed of, for example, a cemented carbide having a Young's modulus of approximately 600 GPa. The fixed-side support blocks **25**B are formed of, for example, a cemented carbide having a Young's modulus of approximately 400 GPa. The fixed-side support blocks **25**C and **25**D are formed of, for example, an alloy tool steel having a Young's modulus of approximately 200 GPa.

[0041] The Young's modulus of the fixed-side support blocks **25**C and **25**D with the lowest Young's modulus among the fixed-side support blocks **25** is higher than the Young's modulus of the metal sheet M. In the present embodiment, the metal sheet M is made of, for example, a stainless steel having a Young's modulus of approximately **190** GPa.

[0042] The fixed-side forming blocks **28** are made of, for example, a high-speed tool steel having a Young's modulus of approximately **220** GPa. The fixed-side backing plate **23** and the shims **26** are made of, for example, alloy tool steels having Young's moduli of approximately 200 GPa.

Movable Die **30**

[0043] The movable die **30** has a similar configuration as the fixed die **20**.

[0044] Hereinafter, the description of the structure of the movable die **30** may be omitted by assigning reference numerals “3*” obtained by adding “10” to the reference numerals “2*” used to indicate the structure of the fixed die **20**.

[0045] As shown in FIG. **3**, the movable die **30** includes a ram **31**, a movable-side shoe **32**, a movable-side backing plate **33**, a movable-side holding plate **34**, movable-side support blocks **35**, shims **36**, and a movable-side forming section **37**. The movable-side support blocks **35**, the shims **36**, and the movable-side forming section **37** are held in a holding hole **34***a* of the movable-side holding plate **34**.

[0046] The movable-side forming section **37** includes a second forming surface **37***a* that forms the metal sheet M. The second forming surface **37***a* faces the first forming surface **27***a*. The movable-side forming section **37** is formed by movable-side forming blocks **38**, which are arranged side by side while being adjacent to each other in the planar direction. The second forming surface **37***a* is formed by continuous lower surfaces of the movable-side forming blocks **38**.

[0047] The boundaries between the movable-side forming blocks **38** and the boundaries between the fixed-side forming blocks **28** are provided at positions displaced from each other in the planar direction. This prevents the metal sheet M from having unintended steps.

[0048] As shown in FIG. **6**, the movable-side support blocks **35** include one movable-side support block **35**A, two movable-side support blocks **35**B, two movable-side support blocks **35**C, and two movable-side support blocks **35**D.

[0049] As shown in FIGS. **3** and **6**, a portion of the movable-side forming section **37** that forms the

flow passages **102** includes a central portion of the second forming surface **37***a*, and is formed by the movable-side forming blocks **38**. The portion of the movable-side forming section **37** that forms the flow passages **102** is supported by the movable-side support blocks **35**A, **35**B, and **35**C.
[0050] The Young's moduli of the movable-side support blocks **35** increase with increasing proximity to the central portion of the second forming surface **37***a* in the planar direction. In the present embodiment, among the movable-side support blocks **35**, the movable-side support block **35**A, which is closest to the central portion of the second forming surface **37***a*, has the highest Young's modulus. The Young's modulus of the movable-side support blocks **35**B is lower than the Young's modulus of the movable-side support block **35**A and higher than the Young's modulus of the movable-side support blocks **35**C. The Young's modulus of the movable-side support blocks **35**C is the same as the Young's modulus of the movable-side support blocks **35**D.
[0051] The materials of the movable die **30** are the same as the materials of the fixed die **20**. The Young's modulus of the movable-side support block **35**A is thus the same as the
[0052] Young's modulus of the fixed-side support block **25**A. The Young's modulus of the movable-side support blocks **35**B is the same as the Young's modulus of the fixed-side support block **25**B. The Young's modulus of the movable-side support blocks **35**C and **35**D is the same as the Young's modulus of the fixed-side support blocks **25**C and **25**D, and is higher than the Young's modulus of the metal sheet M.
Operation of the Present Embodiment
[0053] The Young's moduli, which indicate the stiffness, of the fixed-side support blocks **25** increase with increasing proximity to the central portion of the first forming surface **27***a* in the planar direction. Consequently, during the press forming of the metal sheet M, the fixed-side support blocks **25** positioned closer to the central portion of the first forming surface **27***a* in the planar direction become progressively more resistant to deformation in the movement direction. Accordingly, the portions of the fixed-side forming section **27** closer to the central portion of the first forming surface **27***a* also become progressively more resistant to deformation in the movement direction.
[0054] Similarly, during the press forming of the metal sheet M, the movable-side support blocks **35** positioned closer to the central portion of the second forming surface **37***a* in the planar direction become progressively more resistant to deformation in the movement direction. Accordingly, the portions of the movable-side forming section **37** closer to the central portion of the second forming surface **37***a* also become progressively more resistant to deformation in the movement direction.
Advantages of the Present Embodiment
[0055] (1) The fixed die **20** includes the fixed-side forming section **27**, which includes the first forming surface **27***a*, and the fixed-side support blocks **25**, which support the fixed-side forming section **27**. The movable die **30** includes the movable-side forming section **37**, which includes the second forming surface **37***a*, and the movable-side support blocks **35**, which support the movable-side forming section **37**. The Young's moduli of the fixed-side support blocks **25** increase with increasing proximity to the central portion of the first forming surface **27***a* in the planar direction. The Young's moduli of the movable-side support blocks **35** increase with increasing proximity to the central portion of the second forming surface **37***a* in the planar direction.
[0056] Since this configuration operates in the above described manner, the pressing pressure is prevented from being insufficient in the central part of the metal sheet M. This prevents uneven distribution of pressing pressure on the metal sheet M. This limits a reduction in the forming accuracy of the press die assembly **10**.
[0057] In a case in which the metal sheet M is made of a stainless steel, the pressing pressure is likely to be increased, for example, as compared with a case in which the metal sheet M is made of a titanium material. In this configuration, even when the pressing pressure is increased, the pressing pressure acting on the metal sheet M is prevented from being uneven. This limits a reduction in the forming accuracy of the press die assembly **10**. [0058] (2) The Young's modulus of the fixed-side

support blocks **25** with the lowest Young's modulus among the fixed-side support blocks **25** is higher than the Young's modulus of the metal sheet M. The Young's modulus of the movable-side support blocks **35** with the lowest Young's modulus among the movable-side support blocks **35** is higher than the Young's modulus of the metal sheet M.

[0059] This configuration prevents, during press forming, the fixed-side support blocks **25** with the lowest Young's modulus among the fixed-side support blocks **25** from being deformed prior to deformation of the metal sheet M. Similarly, this configuration prevents the movable-side support blocks **35** with the lowest Young's modulus among the movable-side support blocks **35** from being deformed prior to deformation of the metal sheet M. This limits a reduction in the forming accuracy of the press die assembly **10**. [0060] (3) The fixed-side support blocks **25** are arranged in the first direction X. The movable-side support blocks **35** are also arranged in the first direction X.

[0061] When the pressing pressure at the central portion of the metal sheet M, which has the long sides and the short sides, is insufficient, variations in the pressing pressure in the long-side direction of the metal sheet M tend to be greater than those in the short side direction.

[0062] In this regard, the fixed-side support blocks **25** and the movable-side support blocks **35** of the above-described configuration are arranged in the first direction X, which is the long-side direction of the metal sheet M during press forming. This prevents the pressing pressure acting on the metal sheet M from being uneven in the first direction X. This limits a reduction in the forming accuracy of the press die assembly **10**. [0063] (4) The portion of the fixed-side forming section **27** that forms the flow passages **102** includes the central portion of the first forming surface **27***a* and is supported by the fixed-side support blocks **25**. The portion of the movable-side forming section **37** that forms the flow passages **102** includes the central portion of the second forming surface **37***a* and is supported by the movable-side support blocks **35**.

[0064] With this configuration, the portion of the fixed-side forming section **27** that forms the flow passages **102** and the portion of the movable-side forming section **37** that forms the flow passages **102** resist deformation in the movement direction. This prevents the pressing pressure from being uneven in the portion of the metal sheet M where the flow passages **102** are formed. This increases the forming accuracy of the separator **100** for a fuel cell.

Modifications

[0065] The above-described embodiment may be modified as follows. The above-described embodiment and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

[0066] As shown in FIG. **7**, the fixed-side support blocks **25**D may be omitted from the fixed-side support blocks **25**. In this case, the fixed-side forming section **27** is supported by the fixed-side support block **25**A, the two fixed-side support blocks **25**B, and the two fixed-side support blocks **25**C, which are arranged in the first direction X. This modification can be applied to the movable-side support block **35** in the same manner. This configuration also achieves the advantages of the above-described items (1) to (4).

[0067] As shown in FIG. **8**, the fixed-side support blocks **25**C and **25**D may be omitted from the fixed-side support blocks **25**. In this case, the fixed-side forming section **27** is supported by the fixed-side support block **25**A and the two fixed-side support blocks **25**B, which are arranged in the first direction X. This modification can be applied to the movable-side support block **35** in the same manner. This configuration also achieves the advantages of the above-described items (1) to (4).

[0068] The metal sheet M is not limited to being used as the base material of the separator **100** and may also be used in various other products.

[0069] Two or more fixed-side support blocks **25** may be arranged in the short-side direction of the metal sheet M during press forming. This modification can be applied to the movable-side support block **35** in the same manner.

[0070] The Young's modulus of the fixed-side support blocks **25** with the lowest Young's modulus among the fixed-side support blocks **25** may be lower than or equal to the Young's modulus of the metal sheet M. In this case, the fixed-side support blocks **25** with the lowest Young's modulus preferably support a portion of the fixed-side forming section **27** that does not plastically deform the metal sheet M. This modification can be applied to the movable-side support block **35** in the same manner.

[0071] The fixed-side support blocks **25** may include a single fixed-side support block **25**A and multiple fixed-side support blocks **25**B that surround the outer circumference of the fixed-side support block **25**A. Further, the fixed-side support blocks **25** may include multiple fixed-side support blocks **25**C that surround the outer circumference of the fixed-side support blocks **25**B. This modification can be applied to the movable-side support block **35** in the same manner.

[0072] The Young's modulus of the movable-side support block **35**A does not necessarily need to be the same as the Young's modulus of the fixed-side support block **25**A. The Young's modulus of the movable-side support blocks **35**B does not necessarily need to be the same as the Young's modulus of the fixed-side support blocks **25**B. The Young's modulus of the movable-side support blocks **35**C and **35**D does not necessarily need to be the same as the Young's modulus of the fixed-side support blocks **25**C and **25**D.

[0073] The fixed-side forming section **27** does not necessarily need to include multiple fixed-side forming blocks **28**. The fixed-side forming section **27** may include a single block. This modification can be applied to the movable-side forming section **37**.

[0074] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. A press die assembly, comprising: a fixed die; and a movable die configured to move toward and away from the fixed die, wherein the press die assembly is configured to press a metal sheet, the fixed die includes: a fixed-side forming section that includes a first forming surface; and multiple metal fixed-side support blocks arranged in a planar direction that is orthogonal to a movement direction of the movable die, the fixed-side support blocks supporting the fixed-side forming section from a side opposite to the first forming surface in the movement direction, the movable die includes: a movable-side forming section that includes a second forming surface that faces the first forming surface; and multiple metal movable-side support blocks arranged in the planar direction, the movable-side support blocks supporting the movable-side forming section from a side opposite to the second forming surface in the movement direction, Young's moduli of the fixed-side support blocks increase with increasing proximity to a central portion of the first forming surface in the planar direction, and Young's moduli of the movable-side support blocks increase with increasing proximity to a central portion of the second forming surface in the planar direction.

2. The press die assembly according to claim 1, wherein the Young's modulus of the fixed-side support block with the lowest Young's modulus among the fixed-side support blocks, and the Young's modulus of the movable-side support block with the lowest Young's modulus among the movable-side support blocks, are each higher than a Young's modulus of the metal sheet.

3. The press die assembly according to claim 1, wherein the metal sheet has a rectangular shape having long sides and short sides, the fixed-side support blocks are arranged in a long-side direction of the metal sheet during press forming, and the movable-side support blocks are arranged in the long-side direction of the metal sheet during press forming.

4. The press die assembly according to claim 1, wherein the press die assembly is configured to press the metal sheet, which is a base material of a separator for a fuel cell, thereby forming groove-shaped flow passages, through which a fluid flows, on a surface of the metal sheet, a portion of the fixed-side forming section that forms the flow passages includes the central portion of the first forming surface and is supported by the fixed-side support blocks, and a portion of the movable-side forming section that forms the flow passages includes the central portion of the second forming surface and is supported by the movable-side support blocks.